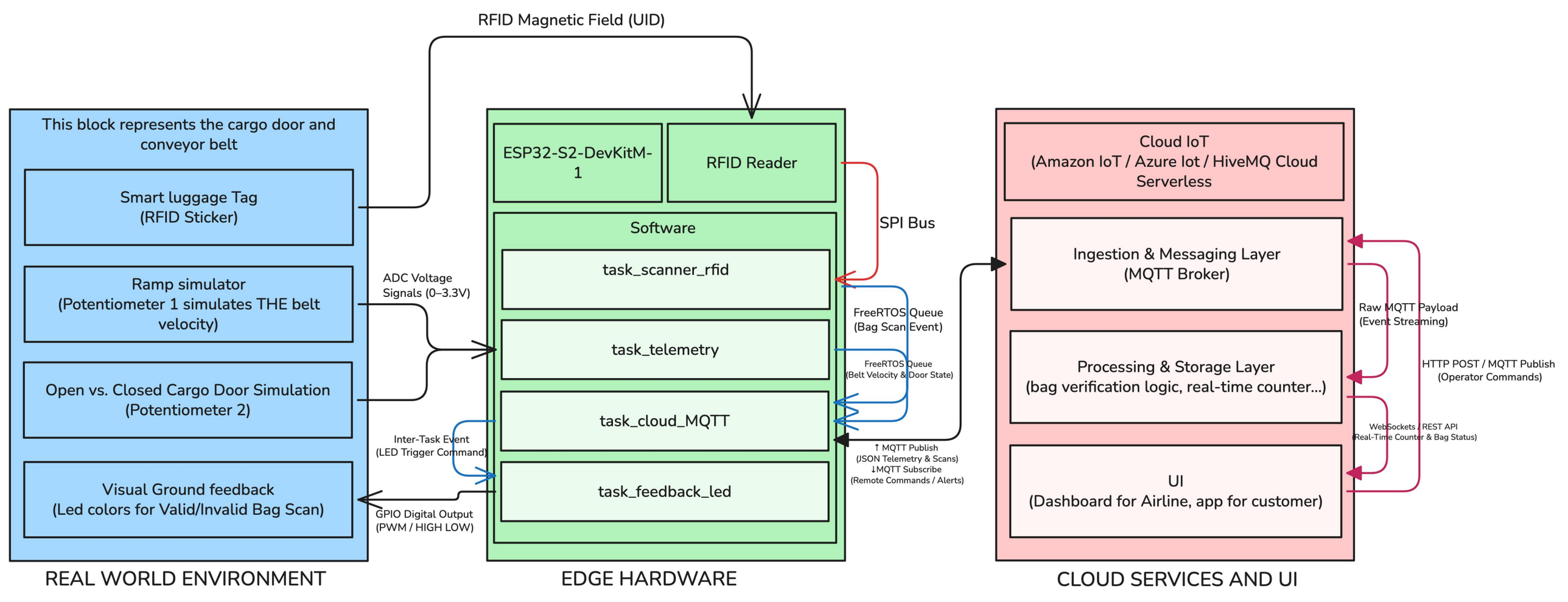

RFID Magnetic Field (UID)
This block represents the cargo door and conveyor belt
Smart luggage Tag (RFID Sticker)
Ramp simulator (Potentiometer 1 simulates THE belt velocity)
Open vs. Closed Cargo Door Simulation (Potentiometer 2)
Visual Ground feedback (Led colors for Valid/Invalid Bag Scan)
ADC Voltage Signals (0–3.3V)
Inter-Task Event (LED Trigger Command)
GPIO Digital Output (PWM / HIGH LOW)
ESP32-S2-DevKitM-1
RFID Reader
Software
task_scanner_rfid
task_telemetry
task_cloud_MQTT
task_feedback_led
SPI Bus
FreeRTOS Queue (Bag Scan Event)
FreeRTOS Queue (Belt Velocity & Door State)
↑ MQTT Publish (JSON Telemetry & Scans)
↓MQTT Subscribe (Remote Commands / Alerts)
Cloud IoT (Amazon IoT / Azure Iot / HiveMQ Cloud Serverless
Ingestion & Messaging Layer (MQTT Broker)
Processing & Storage Layer (bag verification logic, real-time counter...)
UI (Dashboard for Airline, app for customer)
Raw MQTT Payload (Event Streaming)
HTTP POST / MQTT Publish (Operator Commands)
WebSockets / REST API (Real-Time Counter & Bag Status)
REAL WORLD ENVIRONMENT
EDGE HARDWARE
CLOUD SERVICES AND UI